

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3102

3 Credits: 3

Program Core

Source-grounded

# Materials for Printing and

□ To provide basic exposure to the common raw materials and consumables used in

## OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3102 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3102.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Programme            | Diploma in Printing Technology                       |
| Course code          | 3102                                                 |
| Course title         | Materials for Printing and                           |
| Semester             | 3 Credits: 3                                         |
| Course category      | Program Core                                         |
| Periods per week     | 3 (L:2, T:1, P:0) Periods per semester: 45           |
| Periods per semester | 45                                                   |
| Credits              | 3                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

Modules

4 official module sections detected.

Topic entries

22 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

**STUDY METHOD****How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

**Second pass**

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

**Practical evidence**

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

**Revision pass**

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES**

# Objectives, outcomes and mapping

## Course objectives

- To provide basic exposure to the common raw materials and consumables used in
- printing and packaging practices.
- To guide the student to know about various substrates and other essential materials
- used in the printing and packaging industry.
- To understand the general concepts on characteristics, manufacturing and uses of
- primarily used raw materials in printing and packaging.
- To understand the role of primarily used raw materials in printing and packaging in
- deliver quality products at the output.

## Course outcomes

| CO  | Official outcome                                                                                      | Study level           |
|-----|-------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 12 Understanding Summarize the various aspects of paper and                                           | See official syllabus |
| CO2 | 12 Understanding board used in printing and packaging. Explain about various aspects of plastics used | See official syllabus |
| CO3 | 10 Understanding in printing and packaging. Describe about various aspects of                         | See official syllabus |
| CO4 | 9 Understanding miscellaneous printing materials.                                                     | See official syllabus |

## CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 2                      |

| Row | Extracted official mapping |
|-----|----------------------------|
| CO2 | CO2 2                      |
| CO3 | CO3 2                      |
| CO4 | CO4 2                      |

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

| Module   | Official heading                                                          | Topic entries | Hours        |
|----------|---------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Describe about different aspects of printing inks.                        | 4             | As specified |
| Module 2 | packaging.                                                                | 7             | As specified |
| Module 3 | Explain about various aspects of plastics used in printing and packaging. | 5             | As specified |
| Module 4 | Describe about various aspects of miscellaneous printing materials.       | 6             | As specified |

**MODULE 1**

# Describe about different aspects of printing inks.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Describe about different aspects of printing inks.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ Ink-introduction and components

Ink-introduction and components is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Ink-introduction and components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, ink-introduction and components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                           |
|---------------------|---------------------------------------------------------------------------|
| Definition/identity | What Ink-introduction and components means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഇൻക്-ഇൻട്രോഡക്ഷൻ ആൻഡ് കമ്പോണെന്റ്  
 ഇൻക്-ഇൻട്രോഡക്ഷൻ ഡിഫിനിഷൻ/മീനിംഗ് ഇൻക്-ഇൻട്രോഡക്ഷൻ. ഇൻക്-ഇൻട്രോഡക്ഷൻ  
 ഇൻക്-ഇൻട്രോഡക്ഷൻ, steps, application ഇൻക്-ഇൻട്രോഡക്ഷൻ ഇൻക്-ഇൻട്രോഡക്ഷൻ. Technical terms  
 English-ഇൻക്-ഇൻട്രോഡക്ഷൻ ഇൻക്-ഇൻട്രോഡക്ഷൻ.

## ▼ Ink properties

Ink properties is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Ink properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, ink properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                          |
|---------------------|----------------------------------------------------------|
| Definition/identity | What Ink properties means in the official course context |
| Study lens          | Principle → components or stages → result → application  |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഇൻക് പ്രകൃതികൾ നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, steps, application, technical terms English-ഇൻക് പ്രകൃതികൾ.

## ▼ Ink manufacturing

Ink manufacturing is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Ink manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, ink manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                             |
|---------------------|-------------------------------------------------------------|
| Definition/identity | What Ink manufacturing means in the official course context |
| Study lens          | Principle → components or stages → result → application     |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഇൻക് മാനുഫാക്ചറിംഗ് നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ഇ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ **Types of printing inks** 3 **Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and**

Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink,

Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and is an official topic in Module 1 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, types of printing inks 3 understanding ink- introduction, raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. ink properties: optical properties - color, gloss, opacity, transparency. physical properties - length, viscosity, tack, flow, adhesion, scuff resistance. rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. ink manufacturing - flow chart, illustration and description of- varnish manufacture, pigment dispersal, mixing, milling, filtering, quality check, packing and labeling. name of standard ink manufactures in india and world. types of printing inks: offset inks- non-heatset (coldset) inks, heatset inks, quickset inks, sheet-fed inks, gravure inks, flexo inks, screen-printing inks, water-based inks, laser ink, uv ink, process ink colors, high- fidelity ink. specialty ink- non-porous ink, metallic ink, fluorescent ink, magnetic ink, scratch and sniff ink, edible ink, moisture resistant ink, security ink. common ink related printing problems and their remedies-set off, summing, hickies, picking, linting, spot, mottle, strike through. summarize the various aspects of paper and board used in printing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

## Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and |

| Aspect        | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Types of printing inks 3 Understanding Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment

dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High- Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through. Summarize the various aspects of paper and board used in printing and **പേപ്പർ** definition/meaning **പേപ്പർ**. **പേപ്പർ** **പേപ്പർ** **പേപ്പർ**, steps, application **പേപ്പർ** **പേപ്പർ** **പേപ്പർ**. Technical terms English-**പേപ്പർ** **പേപ്പർ**.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 2****packaging.**

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: packaging.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## ▼ 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,

2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding raw materials used. pulping-pulp washing, bleaching, refining, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                  |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                        |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining, definition/meaning. , steps, application . Technical terms English- .

## ▼ Cleaning and Stock preparation. Paper/board making process, Finishing and

Cleaning and Stock preparation. Paper/board making process, Finishing and is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Cleaning and Stock preparation. Paper/board making process, Finishing and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, cleaning and stock preparation. paper/board making process, finishing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Cleaning and Stock preparation. Paper/board making process, Finishing and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നുവെച്ച് Cleaning and Stock preparation. Paper/board making process, Finishing and ഏതെങ്കിലും ഉദാഹരണം definition/meaning എഴുതുക. ഏതെങ്കിലും ഉദാഹരണം, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

### ▼ 3 Understanding Converting, Paper recycling

3 Understanding Converting, Paper recycling is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding Converting, Paper recycling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding converting, paper recycling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                       |
|---------------------|---------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding Converting, Paper recycling means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-3 Understanding Converting, Paper recycling ഫലപ്രസാദം definition/meaning. ഫലപ്രസാദം ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English-ഫലപ്രസാദം ഫലപ്രസാദം.

## ▼ Paper and Board- Properties.

Paper and Board- Properties. is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Paper and Board- Properties. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, paper and board- properties. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                        |
|---------------------|------------------------------------------------------------------------|
| Definition/identity | What Paper and Board- Properties. means in the official course context |
| Study lens          | Principle → components or stages → result → application                |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പേപ്പർ and Board- Properties.

പേപ്പർ എന്നതിന്റെ definition/meaning എഴുതുക. പേപ്പർ ഉണ്ടാക്കുന്ന steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ Paper Sizes. 1

Paper Sizes. 1 is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Paper Sizes. 1 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, paper sizes. 1 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                          |
|---------------------|----------------------------------------------------------|
| Definition/identity | What Paper Sizes. 1 means in the official course context |
| Study lens          | Principle → components or stages → result → application  |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പേപ്പർ സൈസ്. 1 പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-പദപരിവർത്തനം പദപരിവർത്തനം.

## ▼ Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for

Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, paper classifications and their uses. 1 importance of bis & tappi standards for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Paper classifications and their uses. 1 Importance of BIS & TAPPI standards for means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പേപ്പർ പേപ്പർ ക്ലാസ്സിഫിക്കേഷൻ ഉപയോഗം. 1. BIS & TAPPI സ്റ്റാൻഡർഡ്സിന്റെ പ്രാധാന്യം. BIS & TAPPI സ്റ്റാൻഡർഡ്സിന്റെ നിർവ്വചനം/അർത്ഥം. BIS & TAPPI സ്റ്റാൻഡർഡ്സിന്റെ ഉപയോഗം, steps, application. Technical terms English-ൽ പരിഭാഷിക്കുക.

**▼ 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendarizing, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry.**

1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendarizing, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt

sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. is an official topic in Module 2 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 paper & its relation to printing industry. Paper and board
  - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on
    - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendarizing, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 1 paper & its relation to printing industry. paper and board - introduction, invention of papermaking, raw materials used for paper and board making. pulping - definition, classifications-mechanical, chemical, semi - chemical and chemi - mechanical pulping. pulp washing, bleaching, refining, cleaning and stock preparation. paper/board making- overview on - manual method and machines used - cylinder press and fourdrinier machine. manufacturing process - forming, draining, pressing and drying. finishing and converting-calendaring, coating, slitting, cutting, packaging and labeling. mill conditioning. paper recycling-introduction. steps - sorting, deinking, pulping and papermaking. paper and board properties: define - printability & runnability properties. optical properties- color, gloss, opacity, metamerism, brightness, whiteness. structural properties- density, basis weight and grammage, caliper and bulk, wire and felt sides, moisture content and relative humidity, dimensional stability, grain direction, porosity, smoothness. mechanical properties- tear resistance, tensile strength, bursting strength, folding endurance, stiffness, abrasion resistance. paper and board- sizes, classifications and their uses. importance of bis & tappi standards for paper & its relation to printing industry. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - |

| Aspect        | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting- Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 1 paper & its relation to printing industry. Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting- Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill

conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry. പേപ്പറും ബോർഡും തമ്മിലുള്ള വ്യത്യാസം definition/meaning പേപ്പറും ബോർഡും തമ്മിലുള്ള വ്യത്യാസം. പേപ്പറും ബോർഡും തമ്മിലുള്ള വ്യത്യാസം, steps, application പേപ്പറും ബോർഡും തമ്മിലുള്ള വ്യത്യാസം. Technical terms English-ൽ പേപ്പറും ബോർഡും തമ്മിലുള്ള വ്യത്യാസം.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3**

# Explain about various aspects of plastics used in printing and packaging.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Explain about various aspects of plastics used in printing and packaging.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## ▼ 2 Understanding classification

2 Understanding classification is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding classification using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding classification should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                          |
|---------------------|--------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding classification means in the official course context |
| Study lens          | Principle → components or stages → result → application                  |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 2 Understanding classification

പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English- പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം.

## ▼ **Plastics-Manufacturing process-Extrusion.**

Plastics-Manufacturing process-Extrusion. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Plastics-Manufacturing process-Extrusion. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, plastics-manufacturing process-extrusion. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### **Topic study guide**

| <b>Aspect</b>       | <b>What to prepare</b>                                                              |
|---------------------|-------------------------------------------------------------------------------------|
| Definition/identity | What Plastics-Manufacturing process-Extrusion. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്ലാസ്റ്റിക്സ്-Manufacturing process-Extrusion. പ്ലാസ്റ്റിക്സ് എന്നതിന്റെ definition/meaning എഴുതുക. പ്ലാസ്റ്റിക്സ് ഉണ്ടാക്കുന്ന steps, application എഴുതുക. പ്ലാസ്റ്റിക്സ് ഉപയോഗിക്കുന്ന Technical terms English-ൽ എഴുതുക.

## ▼ **Plastics used for printing and packaging.**

Plastics used for printing and packaging. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Plastics used for printing and packaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, plastics used for printing and packaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### **Topic study guide**

| <b>Aspect</b>       | <b>What to prepare</b>                                                              |
|---------------------|-------------------------------------------------------------------------------------|
| Definition/identity | What Plastics used for printing and packaging. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്ലാസ്റ്റിക്സ് ഉപയോഗിച്ചുള്ള പ്രിന്റിംഗ് ആന്റ് പാക്കേജിംഗ്. പ്ലാസ്റ്റിക്സ് ന്റെ definition/meaning എന്താണ്. പ്ലാസ്റ്റിക്സ് ന്റെ ഉപയോഗങ്ങൾ, steps, application എന്തെല്ലാം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

## ▼ Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing,

Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, surface treatment on plastics 2 understanding chemical etching, waxing, varnishing, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എന്നിവിടെ ഉപരിതല ചികിത്സ പ്ലാസ്റ്റിക്‌സിൽ 2  
Understanding Chemical Etching, Waxing, Varnishing, എന്നിവയുടെ നിർവ്വചനം /  
definition/meaning എന്താണ്? . ഏതുവിധത്തിൽ പ്രയോഗിക്കുന്നു , steps,  
application എങ്ങനെയാണ് പ്രയോഗിക്കുന്നത് . Technical terms English-ൽ എഴുതിയിട്ടുള്ള  
അർത്ഥം നൽകുക.

**▼ 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates.**

3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. is an official topic in Module 3 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma

Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 3 understanding laminating, foiling on substrates. plastics - introduction. classification-natural, synthetic, semi-synthetic, thermosetting, thermoplastic. plastics - manufacturing process-extrusion basics. plastics used for printing and packaging - ldpe, hdpe, pp, ps, bopp, pvc, pet, cellophane, plastic corrugated board. surface treatment on plastics - surface cleaning, corona treatment, flame treatment, plasma treatment. chemical etching, waxing, varnishing, laminating, foiling on substrates. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

## Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-3 Understanding Laminating, Foiling on substrates. Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process- Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates. പ്ലാസ്റ്റിക്സ് ന്റെ definition/meaning. പ്ലാസ്റ്റിക്സ് ന്റെ ഉപയോഗങ്ങൾ, steps, application പ്ലാസ്റ്റിക്സ് ന്റെ തരങ്ങൾ. Technical terms English- Malayalam പ്ലാസ്റ്റിക്സ്.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4**

# Describe about various aspects of miscellaneous printing materials.

This module is presented from the official syllabus scope for Materials for Printing and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Describe about various aspects of miscellaneous printing materials.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## ▼ Dampening Solution and Blanket.

Dampening Solution and Blanket. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Dampening Solution and Blanket. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, dampening solution and blanket. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                           |
|---------------------|---------------------------------------------------------------------------|
| Definition/identity | What Dampening Solution and Blanket. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- $\square\square$  Dampening Solution and Blanket.

$\square\square\square\square\square\square\square\square\square$   $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square$   $\square\square\square\square$   $\square\square\square\square\square\square$ , steps, application  $\square\square\square\square$   $\square\square\square\square\square\square\square$   $\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   $\square\square\square\square\square\square\square\square$ .

## ▼ Leather, Al foil, Glass.

Leather, Al foil, Glass. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Leather, Al foil, Glass. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, leather, al foil, glass. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                    |
|---------------------|--------------------------------------------------------------------|
| Definition/identity | What Leather, Al foil, Glass. means in the official course context |
| Study lens          | Principle → components or stages → result → application            |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-പർവ്വതം Leather, Al foil, Glass. പർവ്വതം പർവ്വതം പർവ്വതം പർവ്വതം definition/meaning പർവ്വതം പർവ്വതം പർവ്വതം. പർവ്വതം പർവ്വതം പർവ്വതം, steps, application പർവ്വതം പർവ്വതം പർവ്വതം പർവ്വതം. Technical terms English-പർവ്വതം പർവ്വതം പർവ്വതം പർവ്വതം.

## ▼ Adhesive.

Adhesive. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Adhesive. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, adhesive. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Adhesive. means in the official course context     |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-പ്രശ്നം Adhesive. പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-പദപരിവർത്തനം പദപരിവർത്തനം.

## ▼ Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials &

Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Metals used in printing and packaging. 1 Understanding Photographic and light sensitive materials & using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, metals used in printing and packaging. 1 understanding photographic and light sensitive materials & should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



## ▼ 2 Understanding Image carriers.

2 Understanding Image carriers. is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding Image carriers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding image carriers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                           |
|---------------------|---------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding Image carriers. means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 2 Understanding Image carriers.

പ്രതിബിംബം (image) definition/meaning എന്താണ്. പ്രതിബിംബം സൃഷ്ടിക്കുന്ന മാർഗ്ഗം, steps, application എന്താണ്. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

## ▼ Plate treating solutions & Cleaning solutions. 1 Understanding

**Dampening solution-** purpose of use, ingredients and their properties.

**Blanket - function and types. Adhesive: types, properties and uses.**

**Leather - purpose of use, types. Aluminum foil - properties and**

**application. Glass-** introduction, types, properties and application.

**Metals - types, characteristics and use. Photographic and light**

**sensitive materials - types, features and use. Image Carriers: Cross -**

**sectional view and imaging features of - Lithographic plate, Gravure**

**cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate**

**treating solutions. Cleaning solutions. Text / Reference: T/R Book**

**Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana**

**Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink**

**Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath**

**Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia**

**Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems**

**GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about**

**Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know**

**about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the**

**Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media,**

**Springer, 2002. Online resources: Sl.No Website Link 1.**

**<https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3.**

**<https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5.**

**<https://en.wikipedia.org/wiki/Plastic> 6.**

**[https://en.wikipedia.org/wiki/Substrate\\_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7.**

**<https://www.printweek.in/features/dampening-solutions> 8.**

**<http://printwiki.org/Blanket>**

Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen

printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. <https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3. <https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5. <https://en.wikipedia.org/wiki/Plastic> 6. [https://en.wikipedia.org/wiki/Substrate\\_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/Blanket> is an official topic in Module 4 of Materials for Printing and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging

features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. <https://en.wikipedia.org/wiki/Ink> 2. <http://printwiki.org/Ink> 3. <https://en.wikipedia.org/wiki/Paper> 4. <http://printwiki.org/Pulping> 5. <https://en.wikipedia.org/wiki/Plastic> 6. [https://en.wikipedia.org/wiki/Substrate\\_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/Blanket> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, plate treating solutions & cleaning solutions. 1 understanding dampening solution- purpose of use, ingredients and their properties. blanket - function and types. adhesive: types, properties and uses. leather - purpose of use, types. aluminum foil - properties and application. glass- introduction, types, properties and application. metals - types, characteristics and use. photographic and light sensitive materials - types, features and use. image carriers: cross - sectional view and imaging features of - lithographic plate, gravure cylinder, flexo plate, screen printing stencil, and digital imaging. plate treating solutions. cleaning solutions. text / reference: t/r book title/author t1 tulika das, chemistry in printing, 2nd edition, barnana prakashani, 2011. r.h. leach and r.j. pierce ed, the printing ink manual, 5th edition, kluwer r1 academic

publisher. bishwanath chakravarty, a hand book for printing and packaging, r2 galgotia publications pvt. ltd, 1997. gatf staff solving offset ink problems gatf, 1998 n.r.elred & t. scarlet, what the printer should know about ink, gatf, r3 1995 n.r.elred & t. scarlet, what the printer should know about paper, gatf, r4 1995 r5 n.r.elred & t. scarlet, chemistry for the graphic arts, gatf, 1992. r6 p. kipphan, handbook of print media, springer, 2002. online resources: sl.no website link 1. <https://en.wikipedia.org/wiki/ink> 2. <http://printwiki.org/ink> 3. <https://en.wikipedia.org/wiki/paper> 4. <http://printwiki.org/pulping> 5. <https://en.wikipedia.org/wiki/plastic> 6. [https://en.wikipedia.org/wiki/substrate\\_\(printing\)](https://en.wikipedia.org/wiki/substrate_(printing)) 7. <https://www.printweek.in/features/dampening-solutions> 8. <http://printwiki.org/blanket> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | <p>What Plate treating solutions &amp; Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred &amp; T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred &amp; T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred &amp; T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1. <a href="https://en.wikipedia.org/wiki/Ink">https://en.wikipedia.org/wiki/Ink</a> 2. <a href="http://printwiki.org/Ink">http://printwiki.org/Ink</a> 3. <a href="https://en.wikipedia.org/wiki/Paper">https://en.wikipedia.org/wiki/Paper</a> 4. <a href="http://printwiki.org/Pulping">http://printwiki.org/Pulping</a> 5. <a href="https://en.wikipedia.org/wiki/Plastic">https://en.wikipedia.org/wiki/Plastic</a> 6. <a href="https://en.wikipedia.org/wiki/Substrate_(printing)">https://en.wikipedia.org/wiki/Substrate_(printing)</a> 7.</p> |

| Aspect        | What to prepare                                                                                                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <a href="https://www.printweek.in/features/dampening-solutions">https://www.printweek.in/features/dampening-solutions</a> 8.<br><a href="http://printwiki.org/Blanket">http://printwiki.org/Blanket</a> means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                                                      |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Plate treating solutions & Cleaning solutions. 1 Understanding Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions. Text / Reference: T/R Book Title/Author T1 Tulika Das, Chemistry in Printing, 2nd Edition, Barnana Prakashani, 2011. R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition, Kluwer R1 Academic publisher. Bishwanath Chakravarty, A Hand Book for Printing and Packaging, R2 Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998 N.R.Elred & T. Scarlet, What the Printer should know about Ink, GATF, R3 1995 N.R.Elred & T. Scarlet, What the Printer should know about Paper, GATF, R4 1995 R5 N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts, GATF, 1992. R6 P. Kipphan, Handbook of Print Media, Springer, 2002. Online resources: Sl.No Website Link 1.



**RAPID REVISION****Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**OFFICIAL SELF-LEARNING****Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

**EXAMINATION PREPARATION**

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL****Module-wise questions and answers**

These are practice questions based on official topic coverage, not official examination questions.

**Module 1**

► **Define or explain Ink-introduction and components. (3 marks)**

► **Define or explain Ink properties. (3 marks)**

► **Define or explain Ink manufacturing. (3 marks)**

► **Define or explain Types of printing inks**

**3 Understanding Ink-Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High-Fidelity Ink. Specialty ink- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies- Set off, summing, hickies, picking, linting, spot, mottle, strike through.**

**Summarize the various aspects of paper and board used in printing and. (3 marks)**

## Module 2

► **Define or explain 2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,. (3 marks)**

► **Define or explain Cleaning and Stock preparation. Paper/board making process, Finishing and. (3 marks)**

► **Define or explain 3 Understanding Converting, Paper recycling. (3 marks)**

► **Define or explain Paper and Board- Properties.. (3 marks)**

## Module 3

► **Define or explain 2 Understanding classification. (3 marks)**

► **Define or explain Plastics-Manufacturing process-Extrusion.. (3 marks)**

► **Define or explain Plastics used for printing and packaging.. (3 marks)**

► **Define or explain Surface treatment on plastics 2 Understanding Chemical Etching, Waxing, Varnishing,. (3 marks)**

## Module 4

► **Define or explain Dampening Solution and Blanket.. (3 marks)**

► **Define or explain Leather, Al foil, Glass.. (3 marks)**

► **Define or explain Adhesive.. (3 marks)**

► **Define or explain Metals used in printing and packaging. 1  
Understanding Photographic and light sensitive materials &. (3 marks)**

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Ink-introduction and components**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **2 Understanding raw materials used. Pulping-Pulp Washing, Bleaching, Refining,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **2 Understanding classification**.

3-mark explanation: Explain one principle, process or comparison from this module.

### Module 4

1-mark definition: State the meaning of **Dampening Solution and Blanket..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- Sl.No Website Link <https://en.wikipedia.org/wiki/Ink> <http://printwiki.org/Ink>  
<https://en.wikipedia.org/wiki/Paper> <http://printwiki.org/Pulping>  
<https://en.wikipedia.org/wiki/Plastic>
- [https://en.wikipedia.org/wiki/Substrate\\_\(printing\)](https://en.wikipedia.org/wiki/Substrate_(printing))
- <https://www.printweek.in/features/dampening-solutions>  
<http://printwiki.org/Blanket>

**IN-PAGE SEARCH****Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3102. Verify institutional updates against the current official curriculum.